MILITARY SPECIFICATION

RIVETS, BLIND, NONSTRUCTURAL TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mechanically expanded rivets for blind attachment intended for nonstructural use on military aircraft and other items and components of military equipment.

1.2 Classification. Nonstructural blind rivets shall be furnished in the following classes, as specified (see 6.2):

Class 1 - Universal-head rivets

Class 2 - 100° flush-head rivets

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-H-1581 Hardware (Fasteners and Related Items) Packaging and Packing for Shipment and Storage of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5320

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1312	Fasteners, Test Methods
MS20604	Rivet, Blind, Nonstructural, Universal-Head, Class 1
MS20605	Rivet, Blind, Nonstructural, 100° Flush-Head, Class 2

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. The rivets shall be fabricated from aluminum alloy, nickel-copper alloy (monel), carbon steel, or corrosion-resistant steel, in accordance with the material coding specified on the applicable standards.

3.1.1 Finish. The finishes used on the rivets shall be as specified on the applicable standards. When no finish is required, cadmium or zinc shall not be used as a lubricant. Stem or mandrel, which does not remain in sleeve, shall be lubricated or finished to give corrosion protection while the rivets are in storage.

3.2 Design. The rivets shall consist of two parts: a rivet body and a mandrel or stem. The stem may be either plain or serrated. The rivet shall be installed by pulling the mandrel or stem axially through the rivet body, expanding the shank of the rivet and forming the blind head.

3.2.1 Dimensions. The rivet dimensions shall conform to the dimensions specified on MS20604 and MS20605.

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3.2.2 Roundness and concentricity of heads. Rivet heads shall be round within the limits specified on the applicable standard. They shall be concentric with the shank within the limits specified in table I when examined as specified in 4.4.2.

TABLE I. Tolerances on concentricity of head.

Nominal diameter shank (inch)	Total variation in indicator reading on rivet head	
	Flush head (inch)	Universal head (inch)
0.094, 0.125	0.010	0.010
0.156, 0.187	0.010	0.015
0.250	0.010	0.020

3.3 Expansion. The rivets shall be subjected to the expansion test specified in 4.5.3.

3.4 Strengths. Single shear and tension strengths of the rivets shall be not less than those shown in table II when tested as specified in 4.5.1 and 4.5.2.

TABLE II. Shear and tension strengths.

Material	Min. Single Shear Strength (Lbs)					Min. Tensile Strength (Lbs)				
	Rivet Dia					Rivet Dia				
	.094	.125	.156	.187	.250	.094	.125	.156	.187	.250
Al Alloy 5056	70	125	190	275	490	105	185	285	415	735
Al Alloy 2117	95	195	290	440	770	110	250	390	590	1000
Ni-Cu Alloy (Monel)	105	185	285	415	735	155	270	430	620	1100
Carbon Steel C1008, C1018, C1012	105	185	285	415	735	155	270	430	620	1100
Cres A-286	105	185	285	415	735	155	270	430	620	1100

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3.5 Identification of product. Aluminum-alloy (5056) type nonstructural blind rivets shall be suitably identified by an orange colored dye.

3.6 Workmanship. Rivets shall be of uniform quality and free from seams, fins, clinch or die marks, or other defects which impair their use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test condition specified in 4.3.2.

4.3 Quality conformance inspections. The quality conformance inspections shall consist of examinations and tests as specified in 4.4 and 4.5, respectively.

4.3.1 Sampling.

4.3.1.1 Sampling for examination. Random samples shall be selected from each lot offered to the Government in accordance with MIL-STD-105, Inspection Level I, AQL = 10.0.

4.3.1.2 Lot. An inspection lot shall consist of finished blind rivets which are of the same type and diameter, fabricated by the same process, and produced as one continuous run or order or part thereof.

4.3.1.2.1 A lot of component members of blind rivets, such as sleeves or pins or mechanically expanding blind rivets, shall consist of those components which are of the same type and diameter, fabricated by the same process, and produced as one continuous run or order or part thereof. Inspection already performed on component members need not be repeated after assembly of finished blind rivets submitted as an inspection lot.

4.3.1.3 Rejection and retest. When test failures are due to faulty test procedures, the Inspector is authorized to order retests.

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4.3.1.4 Sampling for tests. Unless otherwise specified, samples shall consist of random samples selected in accordance with table III.

TABLE III. Sampling for mechanical tests.

Lot size	Sample size	Acceptance No.	Rejection No.
Under 5,000	8	1	2
5,000 - 20,000	14	2	3
20,000 - 50,000	20	2	3
Over 50,000	26	3	4

4.3.2 Tests. Each sample rivet selected in accordance with 4.3.1.4 shall be subjected to the following tests:

a. Single shear strength test (see 4.5.1).

b. Tension test (see 4.5.2).

c. Expansion test (see 4.5.3).

4.4 Examinations. Each sample rivet selected in accordance with 4.3.1.1 shall be subjected to the following examinations (4.4.1 and 4.4.2).

4.4.1 Examination of product. All rivets shall be examined to determine conformance to the requirements of this specification with respect to material, workmanship, identification color, instructions, finish, and dimensions. Finish and dimensions shall be checked visually and by means of applicable gages. In case of controversy, gages certified by Government laboratories shall be used.

4.4.2 Roundness and concentricity of head.

4.4.2.1 Universal-head rivets. Roundness and concentricity of universal-head rivets shall be determined by observing the total variation of a dial indicator testing the periphery of the head, as the rivet is rotated with its shank as an axis.

4.4.2.2 100-Degree flush-head rivets. Roundness and concentricity of flush-head rivets shall be determined by observing the total variation of a dial indicator testing the conical part of the head (adjacent to the top of the rivet), as the rivet is rotated with its shank as an axis.

4.5 Test methods.

4.5.1 Single shear test. The rivets shall be subject to the single shear test specified in MIL-STD-1312, Test 20.

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4.5.2 Tension test. The rivets shall be subject to the tensile strength test specified in MIL-STD-1312, Test 8.

4.5.3 Expansion test. The rivets shall be expanded in sheet specimens of suitable thickness. There shall be no splits or cracks in more than 5 percent of the rivets expanded.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. Rivets with the same MS part number shall be preserved in accordance with PPP-H-1581, level A.

5.1.2 Level C. Rivets shall be preserved in accordance with the manufacturer's commercial practice.

5.1.3 Intermediate packaging. Unless otherwise specified by the contracting activity, the quantity of unit packs in the intermediate package shall be at the option of the contractor.

5.2 Packing.

5.2.1 Level A. Rivets preserved in accordance with 5.1.1 shall be packed in accordance with PPP-H-1581, level A, as applicable.

5.2.2 Level B. Rivets preserved in accordance with 5.1 shall be packed in accordance with PPP-H-1581, level B, as applicable.

5.2.3 Level C. The rivets shall be packed in a manner which will insure arrival at destination in a satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Classification rules, or with National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking specified in the contract or order, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This specification is intended to establish and control the quality of blind rivets intended for non-structural use on military aircraft and other items and components of military equipment. For applications involving use of structural-type blind rivets, use MS20600, MS20601, MS20602, or MS20603, as covered by MIL-R-7885.

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. MS part number.

c. Quantity.

d. Levels of preservation and packing.

e. Instructions shall be included in each carton of rivets purchased by the Government to specify:

(1) Use and applications of the rivets.

(2) Setting of the expanding tool to be used (if applicable).

(3) Any special precautions to be observed.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Navy - AS
Air Force - 99

Review activities:
Army - MI, AR
DLA - IS

User activities:
Navy - SH

Preparing activity:
Navy - AS
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